

A Finite-Dimensional Counterexample to a Depth Question in Resonance Index

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Status

Counterexample, likely valid. This answers negatively the Question following Theorem 2.7 in Azamov's *Spectral flow and resonance index* [1]. It does not address the later property A /property B conjectures.

Source Question

The source proves that, for a resonance vector χ , the following are equivalent: $V\chi \perp \mathcal{V}_\lambda$, χ has depth at least one, and

$$\mathbf{A}_\lambda(r_\lambda)S_\lambda\chi = -\chi.$$

It then asks whether these are also equivalent to the condition that $S_\lambda\chi$ is a resonance vector.

2.3. Depth of resonance vectors and depth criteria. Recall that a resonance vector χ has *depth* at least k if there exists a resonance vector χ_1 such that

$$\chi = (1 + (r_\lambda - s)R_\lambda(H_s)V)^k \chi_1.$$

Equivalently, a resonance vector χ has depth at least k if there exists a resonance vector χ_1 such that $\chi = \mathbf{A}_\lambda^k \chi_1$. A resonance vector has depth k if it has depth at least k but not at least $k + 1$.

The following theorem provides a criterion for a resonance vector to have depth ≥ 1 .

Theorem 2.7. *Let χ be a resonance vector. The following assertions are equivalent.*

- (i) *The vector $V\chi$ is orthogonal to $\mathcal{V}_\lambda$.*
- (ii) *The depth of the vector χ is at least 1.*
- (iii) *The equality $\mathbf{A}_\lambda(r_\lambda)S_\lambda\chi = -\chi$ holds.*

Proof. (i) $\Rightarrow$ (iii). By (1.8), the meromorphic operator-function $A_\lambda(s)$ has the following Laurent expansion at $s = r_\lambda$:

$$(2.22) \quad A_\lambda(s) = \tilde{A}_\lambda(s) + \sum_{j=0}^{d-1} (s - r_\lambda)^{-j-1} \mathbf{A}_\lambda^j,$$

where $\mathbf{A}_\lambda^0 = P_\lambda$. From this we obtain

$$1 + (r_\lambda - s)A_\lambda(s) = (r_\lambda - s)\tilde{A}_\lambda(s) + (1 - P_\lambda) - \sum_{j=1}^{d-1} (s - r_\lambda)^{-j} \mathbf{A}_\lambda^j.$$

Applying the latter series to the vector $S_\lambda\chi$ and retaining only the term $(s - r_\lambda)^{-1}$ gives

$$(1 + (r_\lambda - s)A_\lambda(s))S_\lambda\chi = \dots - (s - r_\lambda)^{-1} \mathbf{A}_\lambda S_\lambda\chi + \dots$$

Since χ is a resonance vector, the former series applied to the vector χ also gives

$$A_\lambda(s)\chi = \dots + (s - r_\lambda)^{-1}\chi + \dots$$

Since $V\chi \perp \mathcal{V}_\lambda$, according to Theorem 2.6 the last two Laurent expansions are equal and therefore $\mathbf{A}_\lambda S_\lambda\chi = -\chi$.

(iii) $\Rightarrow$ (ii). This is obvious.

(ii) $\Rightarrow$ (i). Let χ be a resonance vector of depth at least one. By definition of depth, there exists a resonance vector χ' such that $\mathbf{A}_\lambda\chi' = \chi$. Hence, for any eigenvector $\varphi \in \mathcal{V}_\lambda$ we have

$$\langle V\chi, \varphi \rangle = \langle V\mathbf{A}_\lambda\chi', \varphi \rangle = \langle V\chi', \mathbf{A}_\lambda\varphi \rangle = 0,$$

where the last equality holds since $\mathbf{A}_\lambda(r_\lambda)$ eliminates any eigenvector. □

Question. Are the items of this theorem also equivalent to this one: (iv) $S_\lambda\chi$ is a resonance vector? The implication (iii) $\Rightarrow$ (iv) is obvious, so the question is whether (iv) $\Rightarrow$ (iii)?

Idea of the Proof

The question is local and finite-dimensional. For a straight-line path $H_s = H_0 + sV$, resonance vectors are the range of the Laurent idempotent P_0 in the expansion of

$$A_0(s) = (H_0 + sV)^{-1}V$$

at the pole $s = 0$. Thus one can disprove the implication by finding a vector $\chi \in \text{im } P_0$ such that $S_0\chi \in \text{im } P_0$, but $(\mathbf{A}_0 S_0 + I)\chi \neq 0$. The example below is a 5×5 real symmetric matrix calculation.

Counterexample

Counterexample. *The condition that $S_\lambda \chi$ is a resonance vector does not imply $\mathbf{A}_\lambda(r_\lambda)S_\lambda \chi = -\chi$.*

Proof. Work on $\mathbb{C}^5$, put $\lambda = 0$, $r_\lambda = 0$, and let

$$H_0 = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad V = \begin{pmatrix} 0 & -1 & -1 & 0 & 1 \\ -1 & 0 & -1 & 1 & 1 \\ -1 & -1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{pmatrix}.$$

Both matrices are self-adjoint. In finite dimension the perturbation is relatively compact, and 0 is outside the essential spectrum. Moreover

$$\det(H_0 + sV) = 3s^4(s - 1),$$

so $s = 0$ is an isolated resonance point for the line $H_s = H_0 + sV$.

With respect to $\mathbb{C}^5 = \widehat{\mathcal{H}} \oplus \mathcal{V}_0$, where $\mathcal{V}_0 = \text{span}\{e_4, e_5\}$, the source's operator $S_0 = R_0(\widehat{H}_0)V$ is

$$S_0 = \begin{pmatrix} 0 & -1 & -1 & 0 & 1 \\ -1 & 0 & -1 & 1 & 1 \\ 1 & 1 & -1 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

An exact Laurent expansion of $A_0(s) = (H_0 + sV)^{-1}V$ at $s = 0$ is

$$A_0(s) = \widetilde{A}_0(s) + s^{-1}P_0 + s^{-2}\mathbf{A}_0 + s^{-3}\mathbf{A}_0^2,$$

where

$$P_0 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ -2 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

and

$$\mathbf{A}_0 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & -\frac{1}{3} & -\frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} & 0 & 0 \\ 0 & -\frac{2}{3} & \frac{1}{3} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Here $P_0^2 = P_0$, $\mathbf{A}_0^2$ is the coefficient of s^{-3} , and $\mathbf{A}_0^3 = 0$, as required for the Laurent data of the resonance point.

Now set

$$\chi = (0, 1, 0, -2, 1)^T.$$

Then

$$P_0\chi = \chi,$$

so χ is a resonance vector. Also

$$S_0\chi = (0, -1, 2, 0, 0)^T$$

and

$$P_0 S_0 \chi = S_0 \chi,$$

so $S_0 \chi$ is a nonzero resonance vector. However,

$$(\mathbf{A}_0 S_0 + I) \chi = (0, \frac{2}{3}, \frac{1}{3}, -\frac{2}{3}, 1)^T \neq 0.$$

Equivalently,

$$\mathbf{A}_0 S_0 \chi \neq -\chi.$$

Thus condition (iv) in the source question holds while item (iii) of Theorem 2.7 fails. This disproves the proposed equivalence. $\square$

Verification Notes

The finite-dimensional computation is reproduced in `code/verify_counterexample.py`. The script recomputes the Laurent coefficients directly from the full inverse $(H_0 + sV)^{-1}V$ using exact rational arithmetic, checks $P_0^2 = P_0$, $\mathbf{A}_0^2$ and $\mathbf{A}_0^3 = 0$, and verifies the displayed identities for χ .

Novelty Check and Limitations

Cheap run indexes contained no prior packet for arXiv:1604.07006. Web searches on 2026-07-19 for the exact source question, $S_\lambda \chi$ resonance-vector phrases, and the property A /property B conjecture found the source paper and adjacent spectral-flow material, but no later explicit answer to this local Question.

This packet only answers the Question after Theorem 2.7. It does not disprove the later conjectures that property B implies property A , that property B holds, or that property A holds.

Human Review Recommendation

Accept as a finite-dimensional counterexample to the local Question after Theorem 2.7. Reviewers should check the Laurent-coefficient convention against equation (2.22) of the source, which is exactly how the verifier constructs P_0 and $\mathbf{A}_0$.

References

- [1] N. Azamov, *Spectral flow and resonance index*, arXiv:1604.07006v2, 2016.